

TITLE OF MY THESIS ON A TOPIC OF AN INTERESTING AREA OF ENGINEERING

STUDENT NAME 1

M.Sc. ENGINEERING THESIS



**DEPARTMENT OF ELECTRICAL, ELECTRONIC AND
COMMUNICATION ENGINEERING
MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY
DHAKA, BANGLADESH**

JANUARY 2023

SURNAME1

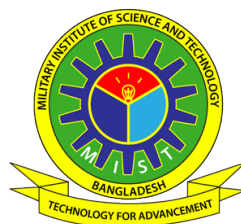
M.Sc. ENGINEERING THESIS

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STUDENT NAME 1 (SN. XXXXXXXXXXXX)

A Thesis Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Electrical, Electronic and
Communication Engineering



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M.Sc. Engineering Thesis

By

STUDENT NAME 1 (SN. XXXXXXXXXXXX)

Approved as to style and content by the Board of Examination on 22 January 2023 :

Lieutenant Colonel Hussain Md. Abu Nyeem, PhD, EME
Instructor Class A
Department of EECE, MIST , Dhaka, Bangladesh

Chairman
(Supervisor)
Board of Examination

Md Golam Mostafa, PhD
Professor
Department of EECE, MIST, Dhaka, Bangladesh

Member
(Internal)
Board of Examination

Brigadier General Md Rezaul Awal, psc
Head of the Department
Department of EECE, MIST, Dhaka, Bangladesh

Member
(Ex-officio)
Board of Examination

Dr. Satya Prasad Majumder
Professor
Department of EEE, BUET, Dhaka-1205

Member
(External)
Board of Examination

Department of Electrical, Electronic and Communication Engineering
MIST, Dhaka

TITLE OF MY THESIS ON A TOPIC OF AN INTERESTING AREA OF ENGINEERING

DECLARATION

I hereby declare that the study reported in this thesis entitled as above is my own original work and has not been submitted before anywhere for any degree or other purposes. Further, I certify that the intellectual content of this thesis is the product of my own work and that all the assistance received in preparing this thesis and sources have been acknowledged and cited in the reference section.

Student Name 1

Department of Electrical, Electronic and Communication Engineering
MIST, Dhaka

TITLE OF MY THESIS ON A TOPIC OF AN INTERESTING AREA OF
ENGINEERING

DEDICATION

A Thesis

By

Student Name 1

Dedicated to my parents for supporting and encouraging
me to believe in myself.

ACKNOWLEDGEMENTS

Foremost, I would like to express my sincere gratitude to my supervisor, Lieutenant Colonel Hussain Md. Abu Nyeem, PhD, EME of the Department of EECE, MIST for the useful comments, remarks and engagement through the learning process of this master thesis. I always received appropriate guidelines,

I would also like to thank I would also like to extend my heartfelt gratitude to I also thank the

Finally, I must express my profound gratitude to

ABSTRACT

Title of My Thesis on a Topic of an Interesting Area of Engineering

Write the abstract of your work as an independent part of your thesis capturing the research problem and its significance, your contributions and its validation, and conclusions.

সারসংক্ষেপ

Detection and Classification of Sensitive Audio-Visual Content for Automated Film Censorship and Rating

স্বয়ংক্রিয় ফিল্ম সেন্সরশিপ এবং রেটিং (AFCR) সম্প্রতি একটি প্রধান মেশিন লার্নিং (ML) গবেষণা ক্ষেত্র হিসাবে আবির্ভূত হয়েছে। অডিওভিজুয়াল সামগ্রীর ব্যাপক বিস্তার এবং বিতরণ ম্যানুয়াল সেন্সরশিপ এবং রেটিংকে জটিল করে তোলে। এইভাবে, মেশিন লার্নিং-ভিত্তিক AFCR ডিজাইন পদ্ধতি আবির্ভূত হয়েছে। প্রথমত, AFCR সিস্টেমের একটি "পরিপূর্ণ" ধারণামূলক মডেল তৈরি করতে হবে, যা সম্ভাব্য ক্লাস এবং তাদের সম্পর্কিত মানদণ্ডের সাথে পরিপূর্ণ। এই কাগজটির লক্ষ্য AFCR সিস্টেমের মানদণ্ডকে প্রমিত করা। বিদ্যমান মেশিন লার্নিং মডেলগুলির তুলনা একটি দক্ষ AFCR সিস্টেম বিকাশে অনুক্রমিক এবং মাল্টিমোডাল বিশ্লেষণের কার্যকারিতা দেখিয়েছে। এই থিসিসটি BrutNet নিয়েও আলোচনা করে, যা সহিংসতা শনাক্ত করার জন্য একটি নতুন ডিপ কনভোলুশনাল নিউরাল নেটওয়ার্ক (DCNN) মডেল। GRU গুলি 160 x 90 এবং 3 সেকেন্ড বা তার বেশি সময়কালের একটি ভিডিও বা ভিডিও ক্লিপের একাধিক ফ্রেম জুড়ে প্যাটার্নে কাজ করে। চিত্রের বৈশিষ্ট্যসমূহ এবং প্যাটার্ন প্রাপ্তির জন্য, টাইম ডিস্ট্রিবিউটেড স্তরের প্রতিটি ফ্রেম বিবেচনা করা হয়েছিল। মডেলটি চৌমাত্রিক থেকে দ্বিমাত্রিক ডেটাতে রূপান্তর করে এবং প্রতি ফ্রেমে 512টি বৈশিষ্ট্য অনুসন্ধান করে। এক-মাত্রিক ভেক্টর তারপর বিভিন্ন ডেস স্তর মাধ্যমে প্রক্রিয়াজাত করা হয়। ফলস্বরূপ, বিষয়বস্তু হয় সহিংস বা অসহিংস। হিডেন এবং আউটপুট স্তরগুলিতে ReLu এবং সিগময়েড অ্যাক্টিভেশন ফাংশন ছিল। মডেলটিকে Google Colab-এ NVIDIA Tesla K80 GPU-তে প্রশিক্ষণ দেওয়া হয়েছিল। প্রশিক্ষিত মডেলটি পূর্ববর্তী LSTM-ভিত্তিক ResNet50 মডেলকে, 90% নির্ভুল পরীক্ষণ সক্ষমতা অর্জন করে ছাড়িয়ে গেছে।

LIST OF NOTATIONS

I	An input image
$\hat{I}$	An embedded version of the input image, I
M	Number of pixels in a row: $M \in \mathbb{N}$
N	Number of pixels in a column: $N \in \mathbb{N}$
NEW	ADD MORE, ADD YOURS

LIST OF ABBREVIATIONS

LSB	Least Significant Bit
MSB	Most Significant Bit

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CHAPTER 1

INTRODUCTION

1.1 Introduction

1.2 Literature Review

1.3 Research Motivation

1.4 Research Objectives

In light of the identified gap in the area of ..., the research presented in this thesis sets its primary goal to develop a ... as mentioned in the previous section. To carry out the project, the specific objectives of this work are outlined as follows.

- a) To investigate the ...
- b) To analyze the ...
- c) To develop algorithms of ...
- d) To develop an experimental setup for the ...

The expected outcome of this work is, therefore, a new ...

1.5 Organization of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 captures the background of the proposed research ...

Chapter 3 presents the proposed development of ...

Chapter 4 presents the evaluation of the proposed scheme for ...

Chapter 5 presents the conclusion of the thesis with a summary of the original contributions and future work.

CHAPTER 2

RELATED WORKS

2.1 Introduction

2.2 Method-I

A few references of earlier work is cited here [1, 2] to generate some entries in the Bibliography section.

2.3 Method-II

2.4 Chapter Summary

CHAPTER 3

PROPOSED SCHEME/METHOD/OWN WORK

3.1 Introduction

3.2 New Section Heading

3.2.1 Subsection-I heading

3.2.1.1 Subsubsection-I heading

3.2.2 Subsection-II heading

3.3 New Section Heading

3.4 Chapter Summary

CHAPTER 4

RESULTS AND DISCUSSIONS

- 4.1 Introduction**
- 4.2 Evaluation Metrics**
- 4.3 Performance Analysis**
- 4.4 Chapter Summary**

CHAPTER 5

CONCLUSIONS AND FUTURE WORKS

5.1 Conclusions

5.1.1 Research outcomes

5.1.2 Research significance

5.2 Future Works

LIST OF PUBLICATIONS

Journal Papers:

- (i) **Authors-Lastname, F. N.** and Others-Lastname, F. N., “Article title,” *Name of the Journal*, Publisher, 2019.
- (ii) **Authors-Lastname, F. N.** and Others-Lastname, F. N., “Journal-Article title,” *Name of the Journal*, Publisher, 2019. (under review)

Conference Papers:

- (iii) **Authors-Lastname, F. N.** and Others-Lastname, F. N., “Conference paper title” *Proceedings title*, City of the conference, Country, Publisher, year, pp. xx-xx.
- (iv) **Authors-Lastname, F. N.** and Others-Lastname, F. N., “Conference paper title” *Proceedings title*, City of the conference, Country, Publisher, year, pp. xx-xx.
- (v) **Authors-Lastname, F. N.** and Others-Lastname, F. N., “Conference paper title” *Proceedings title*, City of the conference, Country, Publisher, year, pp. xx-xx.

REFERENCES

- [1] G. H. Granlund and H. Knutsson, *Signal processing for computer vision*. Springer Science & Business Media, 1994.
- [2] R. A. Jarvis, “A perspective on range finding techniques for computer vision,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, no. 2, pp. 122–139, 1983.

APPENDIX A

MATLAB CODES

A.1 An Example of MATLAB Script

```
1  clc
2  clear all
3  close all
4
5  %% Define the path of the test image for batch run
6  path = strcat(cd, '\TestImage\');
7  pathresult = strcat(cd, '\sResult\SIPIfull\');
8  contents = dir(path);
9
10 Method = [];
11
12 for F=1: numel(contents)-2
13     filename = contents(F+2).name;
14     I= imread(filename);
15
16     load eqdata % the data bits to be embedded
17     Io= double(I(:,:,:)); % original test input image
18     Io(Io==0)=1;
19     Io(Io==255)=254;
20
21     [M, N] = size(Io);
22     pxs =0;
23     pdx =1;
24
25     tic
26     Iw = Io;
27     x=1;
28     for i=1:1:M
29         for j=1:3:N-2
30             if pdx +1 ≤ numel(eqdata)
31                 [ Iw(i,j:j+2), Iems(x:x+2, :), unusedbit, pshifted] = ...
32                     jungembnew(Io(i,j:j+2), eqdata, pdx);
33                 pxs = pxs + pshifted;
34                 pdx = pdx+2-unusedbit;
35                 x=x+3;
36             end
37         end
38     end
39 end
```

A.1.1 An Example MATLAB Function: *MyFunction*(·)

```
1  function [Iwfull, Ctot] = MyFunction(Io, eqdata)
2  Io = double(Io);
```

```
3
4 depth = ceil(log2(double(max(Io(:)+1))));
5 MAX = 2^depth -1;
6 Io(Io==0) = Io(Io==0)+1;
7 Io(Io==MAX) = Io(Io==MAX)-1;
8 k=mod(size(Io),3);
9 Iop = Io(1:end-k(1), 1:end-k(2));
10 pdx =1;
11 [M, N] = size(Iop);
12
13 Idx = double(reshape(1:M*N, [M,N]));
14 %Block-wise zigzag scanning
15 Izdx = im2col(Idx,[3 1],'distinct');
16 Iopz = Iop(Izdx);
17
18 pdx =1;
19 [R, C] = size(Iopz);
20 Iw = double(zeros(R, C));
21 for j = 1:C
22     pix(1:3) = Iopz(:,j);
23     [Iw(:,j), unusedbit] = jungemb(pix, eqdata, pdx);
24     pdx = pdx+2-unusedbit;
25 end
26
27 temp=sortrows([Iw(:), Izdx(:)],2);
28 Iwnew= reshape(temp(:,1), [M, N]);
29
30 Iwfull = uint8(Io);
31 Iwfull(1:end-k(1), 1:end-k(2))=Iwnew(:,:);
32 Ctot= pdx-2+unusedbit-1; %new line
33 end
```